

Roll No.

TME-814

B. TECH. (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

ADVANCED WELDING TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What are different types of flames ? Indicate their applications and chemical reaction thereof. 10 Marks (CO2)

OR

- (b) What is the principle of gas welding ? Explain in brief the process of gas welding. 10 Marks (CO2)

2. (a) What is the function of coated electrode ? Indicate the ingredients used for meeting the different functions of the coated electrodes.

10 Marks (CO1)

OR

- (b) Classify welding processes based on source of energy. 10 Marks (CO1)

3. (a) Compare TIG and MIG welding processes. 10 Marks (CO2)

P. T. O.

(2)

OR

- (b) Describe LASER beam welding process with neat sketch and applications. 10 Marks (CO2)

4. (a) What is the method of storing Acetylene gas in steel cylinders ?

10 Marks (CO1)

OR

- (b) Describe manual arc welding process with a neat sketch.

10 Marks (CO1)

5. (a) Which welding process out of the following has maximum productivity, SAW, CO₂ Welding and TIG welding. 10 Marks (CO1)

OR

- (b) Explain briefly position of welding and different joints used in welding process. 10 Marks (CO1)

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TME-812

B. TECH. (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

NON-CONVENTIONAL ENERGY RESOURCES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What are different types of non-conventional energy resources ?
Explain. 10 Marks (CO1)

OR

- (b) Explain energy exploitation in detail. 10 Marks (CO1)

2. (a) What is energy ? Explain the status of energy on global level.
10 Marks (CO1)

OR

- (b) What do you mean by energy demand analysis ? Explain.
10 Marks (CO1)

3. (a) What is solar constant ? Explain the measurement of solar radiation
with a neat sketch. 10 Marks (CO2)

P. T. O.

(2)

OR

- (b) What are beam radiation, diffused radiation, and global radiation ?
Explain latitude with a neat sketch. 10 Marks (CO2)

4. (a) What is solar collector ? Write energy balance equations for flat plate collector and hence derive the expression for temperature at the outlet of collector. 10 Marks (CO2)

OR

- (b) What is the application of solar collector ? Write energy balance equations for partially covered flat plate collector and hence derive the expression for temperature at the outlet of collector. 10 Marks (CO2)
5. (a) What do you mean by series and parallel connection of solar collectors ?
Explain concentrator collector with a neat sketch. 10 Marks (CO2)

OR

- (b) What is the advantage of using evacuated tubular collector ? Write energy balance equations for evacuated tubular solar collector and hence derive the expression for temperature at the outlet of collector.

10 Marks (CO2)